

Dmitry Prom

AI Engineer · production LLM systems and analytics on conversation data

OPEN TO SENIOR / LEAD APPLIED AI · 5 YEARS IN ENGINEERING, 2 IN AI · REMOTE · UTC+3

xpromx94@yandex.ru
github.com/94spec
94spec.github.io
leadchat.online

SUMMARY

I build production LLM systems that take manual work off revenue teams: a customer conversation becomes verifiable data. And I do the analysis on top of that data – the analysis behind decisions on sales, marketing, product and pricing.

I own the whole path: business policy → model behaviour → architecture → implementation → evaluation → CRM and BI integration → cost and latency → the report people act on. I do not manage people: I work alongside developers and own the AI loop myself.

KEY RESULTS · EVERY NUMBER WITH ITS DENOMINATOR

100% / ≤20% of eligible conversations evaluated, instead of manual sampling denominator – conversations passing the duration and content filter	98% agreement between AI and expert decisions denominator – 500 deals in a frozen reference set (golden set); missed critical violations are checked against a separate threshold	–88% cost of the key LLM scenario per 1,000 deals with output schema, benchmark and quality gates unchanged
45–50% · ×1,2 of routine consultations without an operator; qualified-lead lift denominator – eligible routine consultations; leads – month one vs the previous process	≥ 1 млн ₽ monthly contact-centre saving cost of servicing the same consultation flow, before vs after automation	90 / 464 / 246 production pipelines, checklist criteria, summary fields and detectors plus 55 active checklists and 32 summary configurations

AI quality control across every customer communication

MAJOR EDTECH PLATFORM (RU) · THE WHOLE REVENUE ORGANISATION · BUILT THE LOOP END TO END

Quality control used to listen to a manual sample of calls. Now the communication flow is transcribed, filtered, scored against checklists and turned into structured data for managers, CRM and BI – covering every employee who talks or writes to customers. Behind it: 90 production pipelines, 55 active checklists and 464 scoring criteria. Mine: platform logic, system prompts, model selection, pipelines, scoring rules, delivery and calibration on live traffic.

Evaluation stopped being a sample: 100% of eligible conversations instead of ≤20%.

Analytics that drives decisions

SALES · MARKETING · PRODUCT AND PRICING · COMPETITORS

Tasks arrive from the business as tickets with a goal and an economic rationale. I deliver an answer that can be acted on: what is happening, what supports it and what the data cannot show. Up to 3,000 deals and a six-month horizon, across calls and chats.

- **Why conversion dropped over a quarter.** The month-over-month view showed the problem was not the sales team: traffic was cooling, purchase readiness was falling, money objections were growing. → decisions on traffic buying, script and pricing pitch.
- **"Too expensive" is not always about price.** A two-level reason taxonomy: 'cost' at the top, but inside the group the dominant driver is unavailable instalment financing. That is a payment-instrument problem, not a pricing one.
- **Part of the sales team's load is not sales.** Classifying inbound requests by meaning and by owner surfaced service load inside the sales funnel and a conversion denominator that measured the wrong thing.

Every one of these studies ended in a process change: traffic buying, payment instruments in the offer, a recomputed conversion denominator for the team.

Voice AI

SIPUNI · 10 LINES · REPLACING AN OUTSOURCED CONTACT CENTRE

I turned the contact-centre policy into a live dialogue: script branching, clarifications, barge-in, answers from the knowledge base. Sipuni telephony, 10 concurrent lines, round the clock. Latency budget: 3 seconds to the first reply, up to 10 seconds between replies; it shrinks not by swapping the model but by how early recognition starts and how much context the request carries. Barge-in and branch control live in the system prompt rather than a separate state machine: policy edits ship without a release, but the policy goes through evaluation exactly like code.

45-50% of routine consultations without an operator, ×1.2 qualified leads in month one, saving from 1M RUB per month.

Enterprise RAG

DIFY · 37 DOCUMENTS · 100+ DAILY USERS

Two Dify assistants – one for sales, one for corporate documentation, 37 documents, purely vector retrieval: keyword matching pays off on exact part numbers and rare terms, and on a corpus this size hybrid retrieval would add complexity without a result. I owned chunking, context assembly, grounding and answer quality. The real work is not the model but document preparation – and making the assistant honestly say 'this is not in the documents'.

An answer with a source link in under 10 seconds instead of roughly five minutes of manual search.

LLM evaluation and model economics

4 TASK-SPECIFIC GOLDEN SETS · REGRESSION GATES

Separate sets and metrics for scoring, summaries, semantic analytics and violations. Human calibration, regressions and error analysis after every model change. I moved the same validated output from an expensive model to a compact one: long context and format stability break first, so the release is gated on schema validity and recall on critical violations rather than average accuracy.

98% decision agreement on a frozen 500-deal set; -88% cost of the key scenario.

Lead Chat – my own SaaS

LEADCHAT.ONLINE · PAY PER RESULT: 50 ₪ PER CREATED LEAD

A working product, not a demo: the agent runs a goal-driven dialogue, qualifies the customer, handles objections, answers from the knowledge base and takes actions – books, captures the contact, pushes the lead into CRM. Mine: product concept, AI logic and core architecture. A development partner works with me on the technical side. Next.js, Node.js, tRPC, Prisma, PostgreSQL / pgvector, RAG, tool calling, background jobs, billing.

The product runs in public: live billing, payment per created lead rather than a subscription.

Transcription	A managed speech-recognition service instead of self-hosted Whisper: quality on spontaneous phone speech and diarisation out of the box, predictable price, zero operations. Revisit when volume makes the price gap exceed the cost of operations.
Retrieval	pgvector in the same PostgreSQL instead of a dedicated vector database: search next to the business filters, one query instead of two stores and their sync. Threshold – when retrieval time becomes noticeable next to model time.
Evaluation	An LLM judge against frozen expert labels instead of training a classifier per criterion: criteria change with the policy, and the judge explains its decision. The price is drift when the model version changes, so the set is frozen.
Human in the loop	Where the cost of a mistake is high, the decision stays with a human. A deliberate limit, not an unfinished feature.

THE STANDARD EVERY STUDY IS DELIVERED AGAINST

- Sample control: how many records were included, how many were lost and why – before any percentages.
- A denominator for every share, and the same one when periods are compared.
- Observation and explanation are kept apart: a hypothesis is called a hypothesis and comes with a way to test it.
- Every category is backed by a customer quote, and the stakeholder can verify the labelling by hand.
- A limitations section: what the data does not show and which conclusions it cannot support.

WHAT I CAN OWN FROM SCRATCH

- An LLM quality-evaluation loop – reference set, task-specific metrics, regression gates, model migration without losing quality.
- A voice agent on telephony – dialogue policy, barge-in, a per-stage latency budget, line-capacity sizing.
- An enterprise RAG – document preparation, chunking, context assembly, grounding and an honest refusal.
- An analytics function on conversation data – from a business ticket with a rationale to a decision that was made.

STACK

IN PRODUCTION, UNDER MY RESPONSIBILITY

Python · TypeScript · FastAPI · Node.js · PostgreSQL · pgvector · SQL · Docker · OpenAI API · Anthropic API · GigaChat · YandexGPT · Dify · RAG · AI Agents · Structured Outputs · Tool Calling · LLM-as-a-Judge · Voice AI · CRM / BI · Git

PILOTS AND LAB WORK

LangChain · LangGraph · Qdrant · vLLM · Hugging Face · Fine-tuning / LoRA · RAGAS · A/B Testing · Guardrails · Prompt Injection · NLP

PUBLIC CODE · [GITHUB.COM/94SPEC](https://github.com/94spec) · 7,200 LINES OF SOURCE, 393 TESTS, CI GREEN

llm-evaluation-harness	Dependency-free LLM evaluation framework: four task families, typed error taxonomy, bootstrap intervals, release gates. · 2,300 lines, 125 tests
golden-set-builder	Building reference sets: stratified sampling that reports shortfalls, annotator agreement before any model metric, adjudication without majority vote, freezing with a hash per record. · 1,500 lines, 63 tests
prompt-contracts	Prompts as versioned contracts: output schema, refusal rules, size budget, golden set. Release-impact classification instead of a text diff. · 850 lines, 58 tests
voice-agent-runtime	Voice runtime: dialogue policy as a transition table, barge-in that never resumes, per-stage latency budget against 3 and 10 seconds, line capacity. · 1,100 lines, 73 tests
rag-grounding-lab	What chunking costs, retrieval strategies compared, and sentence-level grounding: an invented number is caught separately, an honest refusal counts as grounded. · 1,200 lines, 54 tests
conversation-intelligence-lab	The same analytics methodology on synthetic data: strict output contract, traceable aggregates, reproducible report. · 300 lines, 20 tests

Every figure here comes with its sample, denominator and comparison rule – happy to walk through the details on a call. Current version of this CV: 94spec.github.io/cv/